

CBSE Test Paper 05
Chapter 12 Thermodynamics

1. Two cylinders A and B of equal capacity are connected to each other via a stopcock. A contains a gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stopcock is suddenly opened. What is the change in internal energy of the gas? **1**
 - a. 10 J
 - b. 50 J
 - c. 100 J
 - d. 0.0 J
2. An ideal gas is taken through a Carnot cycle. The isothermal expansion occurs at 250°C , and the isothermal compression takes place at 50.0°C . Assuming that the gas absorbs 1 200 J of energy from the hot reservoir during the isothermal expansion, find the energy expelled to the cold reservoir in each cycle **1**
 - a. 752 J
 - b. 711 J
 - c. 721 J
 - d. 741 J
3. A steam engine has a boiler that operates at 500 K. The energy from the burning fuel changes water to steam, and this steam then drives a piston. The cold reservoir's temperature is that of the outside air, approximately 300 K. What is the maximum thermal efficiency of this steam engine? **1**
 - a. 40 percent
 - b. 30 percent
 - c. 25 percent
 - d. 45 percent
4. A water heater is operated by solar power. If the solar collector has an area of 6.00 m^2 and the power delivered by sunlight is 550 W/m^2 , how long does it take to increase the temperature of 1.00 m^3 of water from 20.0°C to 60.0°C ? **1**
 - a. 51.7 ks
 - b. 50.7 ks

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- c. 48.7 ks
d. 52.7 ks
5. If C_p and C_v are molar specific heat capacities of an ideal gas at constant pressure and volume respectively and R is the universal gas constant, relation between the three is **1**
- a. $C_p - C_v = 2R$
b. $C_p + C_v = R$
c. $C_p + C_v = 2R$
d. $C_p - C_v = R$
6. Find the values of two molar specific heats of nitrogen. Given, $\gamma = 1.41$ and $R = 8.31 \text{ J mol}^{-1}\text{K}^{-1}$. **1**
7. The internal energy of compressed gas is less than that of the rarefied gas at the same temperature. Why? **1**
8. Explain What is meant by isothermal and adiabatic operations. **1**
9. Calculate the work done for adiabatic expansion of a gas. **2**
10. Why does the density of solid | liquid decreases with rise in temperature? **2**
11. How does Carnot cycle operates? **2**
12. Two bodies at different temperatures T_1 and T_2 are brought in contact. Under what condition, they settle to mean temperature? (after they attain equilibrium) **3**
13. 1 g of water at 100°C is converted into steam of the same temperature. If the volume of steam is 1551 cm^3 , find out the change in internal energy of the water. Latent heat of steam = $2256 \times 10^3 \text{ J/kg}$. Consider atm pressure. **3**
14. State and explain First Law of Thermodynamics. **3**
15. Calculate the work done during the isothermal Process. **5**
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